Virtual City Simulator:

The purpose of the project is to demonstrate how computer science concepts like OOPs, graph, data management, data structure can be applied to solve practical city management problem. (route optimization, traffic management, resource management, emergency response). It has -

- Road Network
- Buildings
- Population
- Vehicles
- Traffic management
- Water and Electricity
- City finance
- Website : A simple web interface can be used to display the simulator.

Core DSA : graph, adjacency list, weighted / undirected graph, queue, priority queue, Dijkstra's algorithm, searching, traversal.

Core OOP : classes, objects, encapsulation, inheritance, constructors and polymorphism.

1. Roads : 1. Graph : The entire road network is a graph
Intersection : vertex
Road = edge

2. Weighted graph : for the distance of the road.

3. Adjacency List : for which intersections are connected.

4. Undirected graph : for two-way road.

5.

```
vector<vector<pair<int,int>>> graph; //adjacency list
void addRoad(int start, int end, int length){
    graph[start].push_back({end,length});
    graph[end].push_back({start,length});
}
```

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2. Buildings : building such as house, hospital, restuarant will be located at the intersection i.e at vertex. The building doesn't have to be the graph node. That could be :

Building ID → connected intersection ID

How adding works?

Building type : house
Building ID : H1
Intersection ID : I1
Population : 5

Population : It will not include hospital / restuarant staff or customers in city population.

Keeping :

Population = people living in house.

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4. Vehicle : initially creating and storing a car with :

Car ID
Car Name

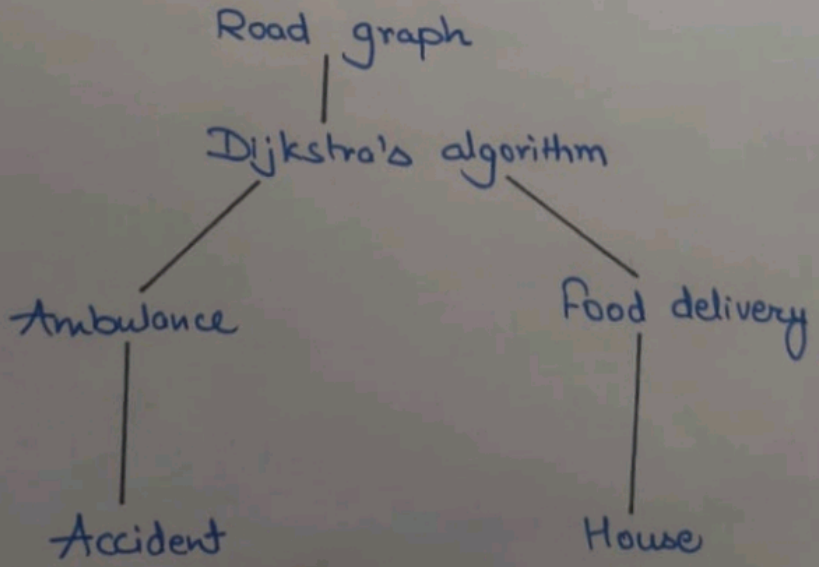
Later can add :

Current location.
Destination.
Route.
Speed.

Vehicle that will run on the road are - car, ambulance, food delivery bike.

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5. For ambulance, food delivery part : Dijkstra's algorithm will be used for both, but for different purposes.



City fund: It manages fund, donations, expenses

The flow is:

Donation
↓
City fund increases
↓
Water / Electricity insufficient?
↓
Purchase resources
↓
City fund decreases

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7. Water and power supply:

Available >= Required
↓
sufficient

Available < Required
↓
insufficient
↓
Use city finance to purchase

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8. Traffic light: It will be placed at intersection where 3 or more roads meet.

- The system will monitor the no. of wating cars on each incoming road.
- It will also track the waiting time of vehicles.
- Priority score will be calculated as No. of Cars X Waiting time
- To prevent starvation, if a road has at least one waiting car for 60 seconds or more, it will receive priority.
- After each signal cycle, priorities will be recalculated dynamically based on current traffic.